INCH-POUND

MIL-DTL-11732F
15 September 2010
SUPERSEDING
MIL-DTL-11732E
25 January 1999

DETAIL SPECIFICATION

CYLINDER, COMPRESSED GAS: CHLORINE; DOT 3A480 OR DOT 3AA480, GENERAL SPECIFICATION FOR

Reactivated after 15 September 2010 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of steel cylinders, with valve and cap, intended for chlorine service.

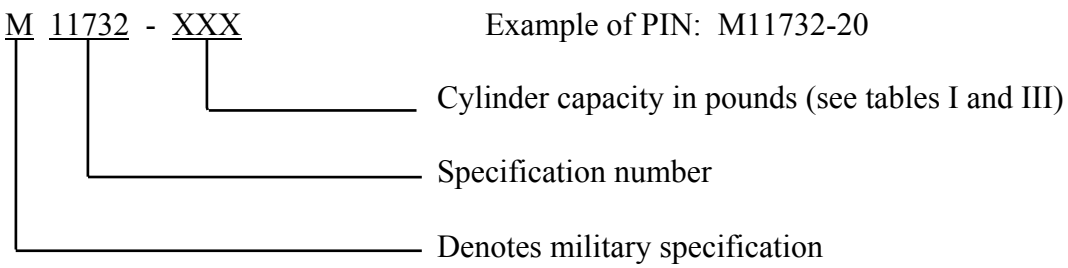
1.2 Classification. The cylinders covered by this specification will be of the following types, as specified (see 6.2).

Type I - 20 pounds
Type II - 30 pounds
Type III - 100 pounds
Type IV - 150 pounds

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1.3 Part or identifying number (PIN). The PIN to be used for gas cylinders acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

TT-C-490	- Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

FEDERAL STANDARD

FED-STD-H28	- Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	- Nitrogen, Technical
A-A-59860	- Valves, Cylinder, Gas (for Compressed or Liquefied Gases)

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-17376	- Caps and Flanges, Compressed-Gas Cylinder, General Specification for

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DEPARTMENT OF DEFENSE STANDARD

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 178.36	- Specification 3A and 3AX Seamless Steel Cylinders
49 CFR 178.37	- Specification 3AA and 3AAX Seamless Steel Cylinders

(Copies of these documents are available online at http://www.gpoaccess.gov/ecfr/ or from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954.)

DRAWING

NAVY

810-1385867	- Decalcomania, for Navy Gas Cylinders

(Copies of drawings required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D2029	- Standard Test Methods for Water Vapor Content of Electrical Insulating Gases by Measurement of Dew Point

(Copies of this document are available from http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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MASTER PAINTERS INSTITUTE

MPI #9	- Alkyd, Exterior Gloss (MPI Gloss Level 6)

(Copies of this document are available online at http://www.paintinfo.com/ or from Master Painters Institute, 2800 Ingleton Avenue, Burnaby, BC V5C 6G7 Canada.)

SSPC: THE SOCIETY FOR PROTECTIVE COATINGS

SSPC-Paint 25	- Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel Type I and Type II
SSPC-SP 5	- White Metal Blast Cleaning
SSPC-SP 6	- Commercial Blast Cleaning
SSPC-SP 10	- Near-White Metal Blast Cleaning

(Copies of these documents are available online at http://www.sspc.org/ or from SSPC: The Society for Protective Coatings, 40 24th Street, Sixth Floor, Pittsburgh, PA 15222.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Configuration. Cylinders shall conform to the requirements of 49 CFR 178.36 for DOT 3A or 49 CFR 178.37 for DOT 3AA, and shall be as specified herein.

3.1.1 Cylinders. The cylinder shall be as specified in table I. All dimensions are in inches. The tolerance for outside diameter (OD) is ±0.5 inch and that for height is ±2 inches.

TABLE I. Cylinder configuration.

Type	Nominal gas capacity (pounds)	Outside dimensions		Min. water capacity (pounds)	Max. tare weight (pounds)
		OD (inches)	Height (inches)		
I	20	6	19	16	25
II	30	6	29	24	35
III	100	8.5	48	80	85
IV	150	10.25	48	120	120

3.1.2 Flanges, valves, and caps. The cylinder shall be supplied with flanges, valves and cap in accordance with 3.4 and figure 1.

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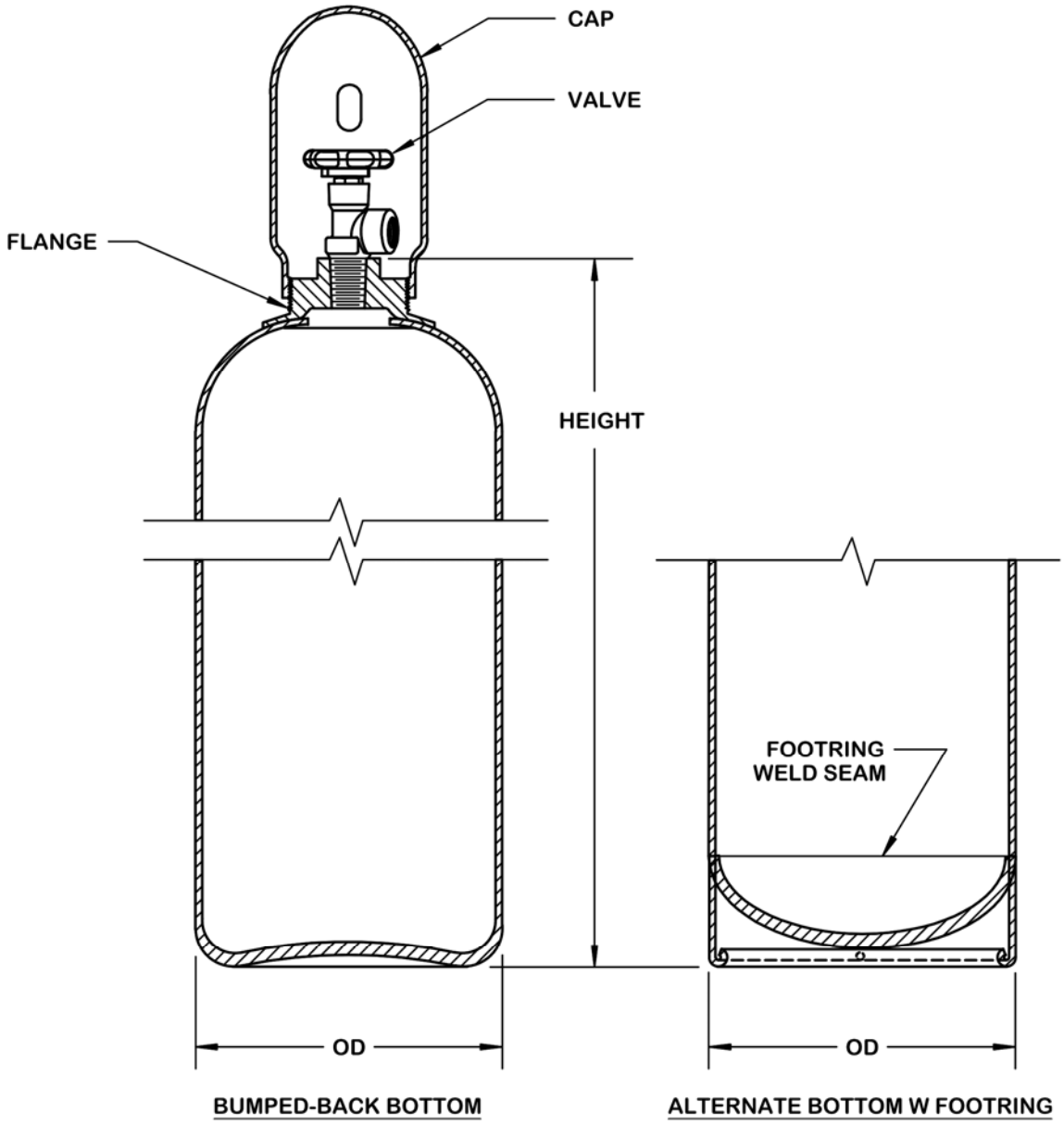


FIGURE 1. Cylinder configuration.

3.1.3 Bottom design. The bottom design shall be bumped back or shall be designed with a foot ring for upright stability. Footrings, when used, shall have a minimum of two ventilation holes and two drainage holes equally spaced.

3.2 Construction. Cylinders shall be true of form and shall be stable when in an upright position. The neck of each cylinder shall be tapped internally with National Gas Taper (NGT) threads: 3/4-14 NGT (Cl)-1. Threads shall conform to FED-STD-H28.

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3.3 Cylinder processing.

3.3.1 Internal surfaces. The internal surfaces of drawn cylinders, prior to necking, shall be free of mill scale, burrs, oxidation products and all foreign matter. When required, only iron-based abrasives shall be used to remove unstable surfaces or oxidation products from the cylinder wall. Metal discoloration, rust bloom, and fixed scale shall not be considered criteria for cylinder rejection.

3.3.2 Cylinder cleaning. After necking, drilling, threading, and deburring, the cylinders shall be cleared of loose particles by vacuum or by inverting the cylinder and applying a compressed air sweep (see 6.3). The cylinder shall be flushed with hot fluids when necessary to remove grease, oil, and fat from processing. Chemicals that will react with chlorine shall not be used.

3.3.3 Cylinder drying. Internal cylinder moisture, from processing or hydrostatic testing, shall be removed by inverting and steaming the surfaces until hot to the touch, followed by drying with filtered air having a dew point less than 30 °F until the dew point of effluent gas is less than 40 °F when tested in accordance with ASTM D2029 or by equivalent method. In lieu of filtered air, drying by suction of ambient air through the cylinder is allowed. The residual air in the dry cylinder shall be replaced by introducing dry nitrogen gas conforming to A-A-59503, type I, grade B, through a bayonet extending to the bottom end of the cylinder. Closure shall be carried out immediately to limit atmospheric interchange.

3.4 Closure. Cylinders shall be equipped with a flange and a cap conforming to MIL-DTL-17376. A plug or closed valve conforming to A-A-59860, as specified, shall be installed in the neck of the cylinder (see 6.2).

3.5 Tare weight. The tare weight of the cylinder shall be the weight of the cylinder, flange, and valve, but exclude the valve protection cap. The tare weight shall be expressed in pounds and ounces expressed in 4-ounce increments to the nearest quarter-pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-). Example: A tare weight marking of "119-4" indicates a weight of 119 pounds, 4 ounces.

3.6 Test date. The cylinders shall be delivered within 1 year of the test date stamped on the cylinder.

3.7 Cylinder identification. Marking shall be in characters not less than 0.25 inch in height.

3.7.1 Standard markings. Standard markings shall be plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck as follows:

a. "DOT-3A480" or "DOT-3AA480" in characters not less than 0.375 inch in height for cylinders greater than 6 inches in outside diameter.
b. Serial number (see 6.5), as registered with the Bureau of Explosives, in characters not less than 0.5 inch in height for cylinders greater than 6 inches in outside diameter.

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c. "US GOVT", Government symbol, as registered with the Bureau of Explosives, in characters not less than 0.5 inch in height for cylinders greater than 6 inches in outside diameter.
d. The inspector's official mark.
e. Date of test, stamped diametrically opposite the above marking.
f. Manufacturer's or re-tester's mark, near the date of test (preferably right above it).
g. The tare weight shall be stamped in a position other than in sequence with the serial number or test date, and preceded by the letters TW.

3.7.2 Additional markings. In addition to the above markings, the cylinders shall bear the following markings:

a. The name of the gas, stenciled on the cylinder in accordance with MIL-STD-101.
b. When specified (see 6.2), two decals conforming to drawing 810-1385867 shall be affixed diametrically opposite to one another, on the sides of the cylinder, 90 degrees from the stenciled name of the gas.

3.8 Treatment and painting.

3.8.1 Cleaning. Each cylinder and cap shall be cleaned in accordance with TT-C-490. Cleaning may be accomplished by:

a. Chemical methods such as solvent cleaning, alkaline cleaning, acid cleaning, pickling, descaling with hydride, or paint stripping; or,
b. Electrochemical cleaning methods such as electrolyte alkaline, electro-polishing, or electrolytic pickling; or,
c. Mechanical means such as blasting, chipping, wire brushing, or grinding.

3.8.1.1 Blasting. Where blasting is appropriate, blast in accordance with SSPC-SP 6, unless SSPC-SP 5 or SSPC-SP 10 is specified (see 6.2). Blasting shall not be used on surfaces that could be damaged by blasting or on components containing such surfaces unless protective masking has been applied.

3.8.2 Treatment. Cylinders shall be treated as soon as possible after cleaning, as specified in 3.8.1, with a zinc phosphate chemical conversion coating in accordance with TT-C-490, type I.

3.8.3 Painting. Coatings shall be applied in an ambient temperature of not less than 50 °F and not more than 90 °F. Paint shall be applied by any method that will deposit the specified film coat thickness and comply with any applicable transfer efficiency regulations. Paint and surface shall be approximately the same temperature except when hot spray is used.

3.8.3.1 Bottom coating. Calcium sulfonate coating, such as Watson Armor Shield or an equivalent, shall be applied to each dry, clean cylinder bottom and footring, when applicable, as soon as practicable after cleaning and treatment (see 3.8.1 and 3.8.2). Dry film thickness of the bottom coating shall be 3.0 to 7.0 one-thousandth of an inch (mil).

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3.8.3.2 Primer. Primer conforming to SSPC-Paint 25 shall be applied to each dry, clean cylinder as soon as practicable after cleaning, treatment, and bottom coating (see 3.8.1, 3.8.2, and 3.8.3.1). Dry film thickness of the primer shall be 0.7 to 1.0 mil.

3.8.3.3 Finishing coat. Finishing coat conforming to MPI #9 shall be applied to each cylinder. The color of the finishing coat shall conform to the cylinder color code requirement of MIL-STD-101. Dry film thickness of the finishing coat shall be 1.0 to 1.5 mils.

3.9 Workmanship. Cylinder, valve, plug, flange, and cap shall be clean and free from grit and loose scale. Edges shall be rounded or chamfered. The cylinder shall be cleaned and free of other surface defects detrimental to the intended use.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as conformance inspection.

4.2 Conformance inspection.

4.2.1 Examination. Each cylinder shall be examined for defects as listed in table II.

TABLE II. Processing defects.

Defect code	Defect	Requirement paragraph
101	Cylinder not in accordance with 49 CFR 178.36 for DOT 3A cylinder or 49 CFR 178.37 for DOT 3AA cylinder	3.1
102	Dimensions not as specified	3.1.1
103	Construction not as specified	3.2
104	Cleaning and drying not as specified	3.3.2, 3.3.3
105	Cylinders not purged with nitrogen	3.3.3
106	Closure not as specified	3.4
107	Tare weight not as specified	3.5
108	Test date more than one year prior to the date of acceptance by the Government	3.6
109	Marking not as specified	3.7
110	Treatment and painting not as specified	3.8
111	Workmanship not as specified	3.9

4.2.2 Tests. Cylinder testing shall be in accordance with the requirements described in 49 CFR 178.36 for DOT 3A or 49 CFR 178.37 for DOT 3AA cylinders (see 3.1). All tests shall be completed within the one-year period prior to the date of acceptance by the Government.

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4.2.2.1 Lot. A lot shall consist of not more than 200 cylinders produced from the same heat of steel, have the same treatment cycle, and be offered for delivery at the same time.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These cylinders are intended for use in the shipment and storage of chlorine. These cylinders may be adapted for shipment and storage of methyl bromide.

6.1.1 Military-unique rationale. These cylinders may be subjected to harsh environments, including temperature extremes and saltwater corrosion. For this reason, they require surface preparation and painting beyond that normally provided for commercial cylinders.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of cylinder required (see 1.2).
c. When a plug or valve is required (see 3.4).
d. When decals are required (see 3.7.2).
e. Blasting specification, if appropriate (see 3.8.1.1).
f. Packaging requirements (see 5.1).

6.3 Loose particulate matter in stored cylinders. Cylinders that have been cleaned, dried, valved or plugged, and subjected to storage and handling within delivery time limits, will be acceptable for chlorine service if accumulated loose ferro-related particulate matter resulting from handling does not exceed 1 gram per 1,500 cubic inches of cylinder capacity.

6.4 Cross-reference. The differences in cataloging data for chlorine cylinders between this revision and MIL-C-11732/1 are listed by capacity in table III.

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TABLE III. Cross-reference.

Nominal gas capacity (pounds)	MIL-C-11732/1	MIL-DTL-11732F
20	C-11732/I-1	M11732-20
30	C-11732/I-2	M11732-30
100	C-11732/I-3	M11732-100
150	C-11732/I-4	M11732-150

6.5 Serial numbering. Cylinder serial numbers will be prefixed by a two or three digit symbol designating the procuring agency and will have a two-letter suffix designating the supplier furnished by the contracting officer. The supplier will assign the serial numbers to the individual cylinders. These numbers may be concurrent with the supplier's regular production numbers or of a series established specifically for the customer's cylinders. However, all cylinders on a given contract will be numbered consecutively and controls will be exercised to preclude duplication with future deliveries to the Government.

6.6 Subject term (key word) listings.

Bottom coating
Calcium sulfonate
Cap
Flange
Footring
Methyl bromide
National Gas Taper
Tare weight
Valve

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - EA
Navy - SH
Air Force - 68
DLA - GS

Review Activities:
Navy - MC, YD

Preparing Activity:
DLA - GS7

(Project 8120-2010-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.daps.dla.mil/.

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